

HUM-SENSITIVE TOUCH ALARM



■ PRADEEP G.

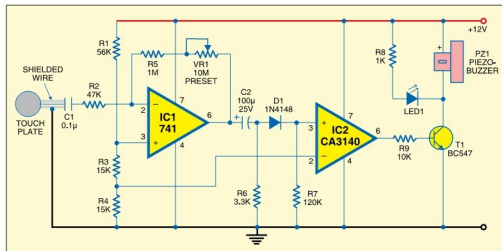
Radiation signals from mains wiring can travel a few metres of distance. These can be induced by the electromagnetic field in the human body also.

This touch-sensitive alarm is based on generation of the AC hum signal. When someone touches the touch plate, low-power AC hum (the same as induced from AC wiring of the house) is generated on the touch plate. This signal is first amplified by the high-

gain preamplifier built around IC 741 (IC1) and then fed to another op-amp CA3140 (IC2) that is wired as a voltage comparator.

When IC2 receives the amplified signal at its input pin 3, its output goes high. As a result, transistor T1 conducts to sound the buzzer. At the same time, LED1 glows. For satisfactory working of this unit, power it from mains-derived 12V.

Assemble the circuit on a general-purpose PCB and enclose in a small cabinet. Use a shielded wire to connect the touch plate to the circuit keeping the length of the wire and the size of the touch plate as short as possible. ●



ELECTRIC BUG ZAPPER

■ T.A. BABU

None of us likes bugs at home. To kill these flying insects, they should first be attracted and then electrocuted. Bug zapper is one such device with a high-voltage

electrocuting circuit and an insect-attracting UV lamp of 365 nm wavelength. This ultraviolet fluorescent lamp is mounted in the middle of the cabinet and a pair of carefully spaced, electrically insulated, charged wire grids surround the light. When an

insect comes close enough to the mesh pair, an electrical arc is formed, the dielectric breaks down and current flows through the insect's body. Electrocuting the insect doesn't require it

to touch both the wires as an arc forms in the air gap over 1800V.

In the schematic, the grid is charged to 1800V by the voltage multiplier built around diodes D1 through D6

